

Run 01 | Fear of failure

Global (including India)

IV only

Target Fear of failure prevents starting a business; binary 1 vs. 0
Sample 1,334,667 respondents **Coverage** 89 countries; 2015–2022
Positive cases 583,346 (43.71%) **Case time** 28m 47s
Train / val. / test 907,575 / 160,159 / 266,933 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.6818). Highest F1: Random Forest (0.6376). The all-negative accuracy baseline is 56.29% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.6818	0.6173	55.81	49.69	88.95	0.6376	4m 36s
XGBoost	0.6706	0.5998	54.85	49.09	89.02	0.6328	26s
CatBoost	0.6656	0.5945	54.67	48.98	88.69	0.6310	1m 42s
H2O GBM	0.6646	0.5918	54.68	48.98	88.75	0.6312	1m 46s
Decision Tree	0.6149	0.5398	43.71	43.71	100.00	0.6083	30s
Logistic Regression	0.6075	0.5287	48.84	45.82	93.40	0.6148	1m 11s
Lasso	0.6073	0.5285	48.81	45.80	93.41	0.6147	4m 03s
Elastic Net	0.6073	0.5285	48.82	45.80	93.38	0.6146	3m 04s
Ridge	0.6073	0.5285	48.82	45.81	93.40	0.6147	7m 26s
Naive Bayes	0.5801	0.5136	44.06	43.82	99.28	0.6081	3m 19s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	5,841.70	6.38
2	Self-efficacy	5,410.60	5.91
3	Good career choice	3,913.54	4.27
4	Education	3,362.47	3.67
5	Opportunity perception	3,283.97	3.58
6	Knowing entrepreneurs	3,203.20	3.50
7	Economic quality	3,122.32	3.41
8	Ease of starting a business	3,053.99	3.33

Share is the percentage of total forest impurity importance across all 49 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	3,362.47	3.67
2	Economic quality	3,122.32	3.41
3	Market access and infrastructure	2,994.57	3.27

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 4 batch, 29 September 2026; 01_obj4_global_iv_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.